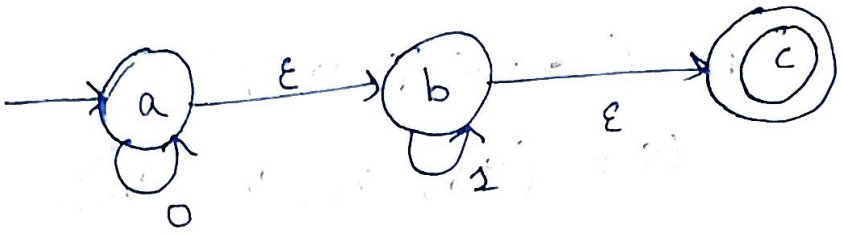
CLASS TEST - 4

Converting NFA-ε to NFA without ε



transition table for NFA-ε

δ	0	1	ε
a	a	∅	b
b	∅	b	c
c	∅	∅	∅

Find epsilon closure for this

epsilon closure for state

ε-closure(a) = {a, b, c}

ε-closure(b) = {b, c}

ε-closure(c) = {c}

transition table for NFA without ε

δ'	0	1
a	{a,b,c}	{b,c}
b	$\emptyset$	{b,c}
c	$\emptyset$	$\emptyset$

To find the transition table for NFA without ε

$$\delta'(a,0) = \varepsilon\text{-closure}(\delta(\varepsilon\text{-closure}(a), 0))$$

$$= \varepsilon\text{-closure}(\delta(\{a,b,c\}, 0))$$

$$= \varepsilon\text{-closure}(\delta(a \cup \emptyset \cup \emptyset))$$

$$= \varepsilon\text{-closure}(a)$$

$$= \{a,b,c\}$$

$$\delta'(a,1) = \varepsilon\text{-closure}(\delta(\{a,b,c\}, 1))$$

$$= \varepsilon\text{-closure}(\emptyset \cup b \cup \emptyset)$$

$$= \varepsilon\text{-closure}(b)$$

$$= \{b,c\}$$

$$\delta'(b,0) = \varepsilon\text{-closure}(\delta(\{b,c\}, 0))$$

$$= \varepsilon\text{-closure}(\emptyset \cup \emptyset)$$

$$= \emptyset$$

$$\delta'(b,1) = \varepsilon\text{-closure}(\delta(\{b,c\},1))$$

$$= \varepsilon\text{-closure}(b \cup \emptyset)$$

$$= \{b,c\}$$

$$\delta'(c,0) = \varepsilon\text{-closure}(\delta(\{c\},0))$$

$$= \emptyset$$

$$\delta'(c,1) = \varepsilon\text{-closure}(\delta(\{c\},1))$$

$$= \emptyset$$

Construction of NFA without ε

